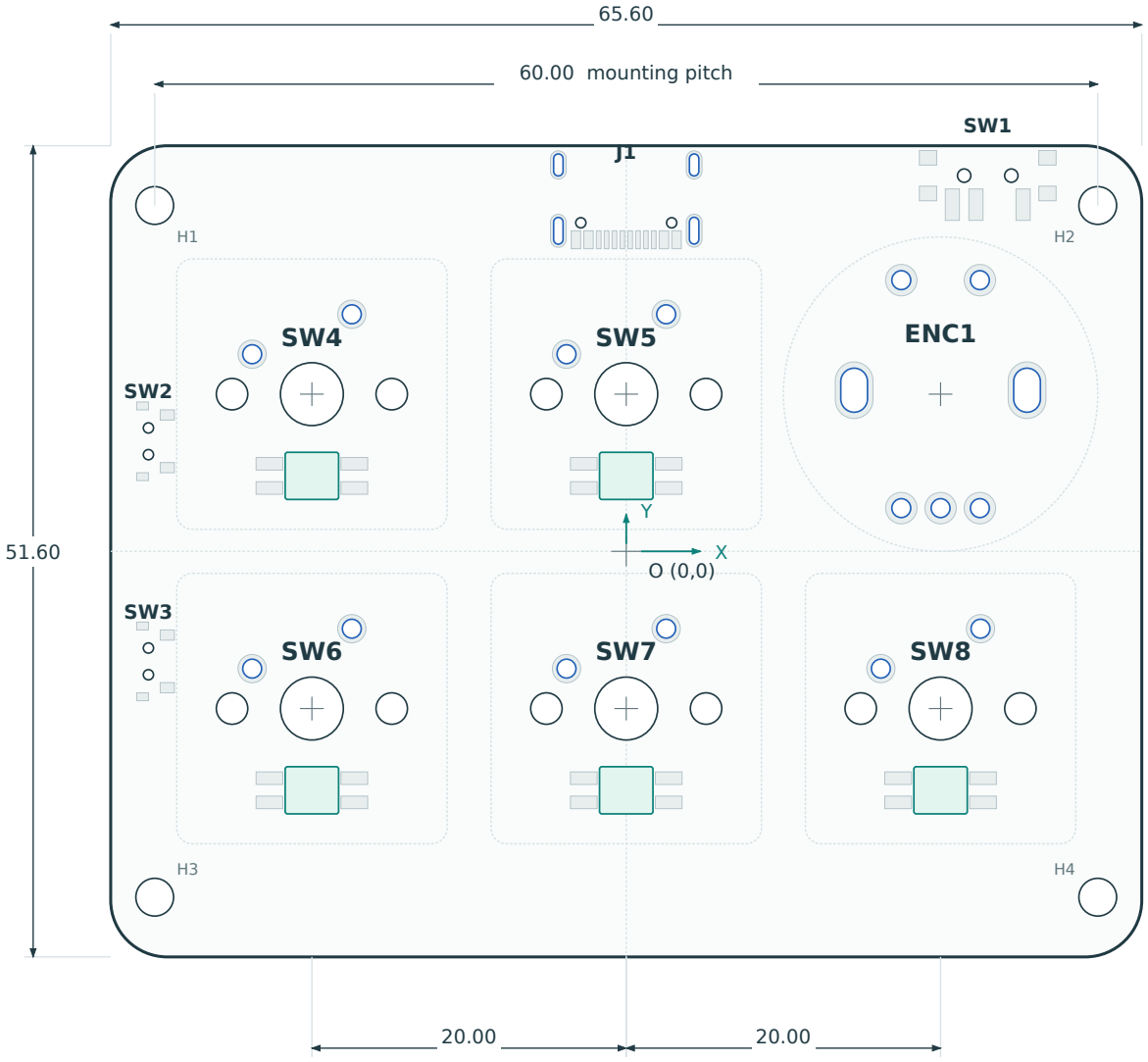
CH592 / PCB mechanical layout

Board 65.6 × 51.6 × 1.6 · R3.6 corners · 20 mm key pitch · Print05 structural proposal



FRONT / USER SIDE · dashed outlines = lid openings / knob envelope

DATUM COORDINATES

Ref.	X	Y
SW4	-20.00	+10.00
SW5	+0.00	+10.00
SW6	-20.00	-10.00
SW7	+0.00	-10.00
SW8	+20.00	-10.00
ENC1	+20.00	+10.00
J1	+0.00	+22.20
SW1	+23.00	+23.90
SW2	-30.40	+7.00
SW3	-30.40	-7.00

H1 / H2: X = ±30, Y = +22

H3 / H4: X = ±30, Y = -22

4 × Ø2.40 NPTH for M2 screws

- PCB mechanical / NPTH hole
- PCB plated hole / slot
- RGB through-board light window
- SW2 / SW3: access through left wall

Board plane dimensions before 6° tilt.

Lower-left bounding-box origin:

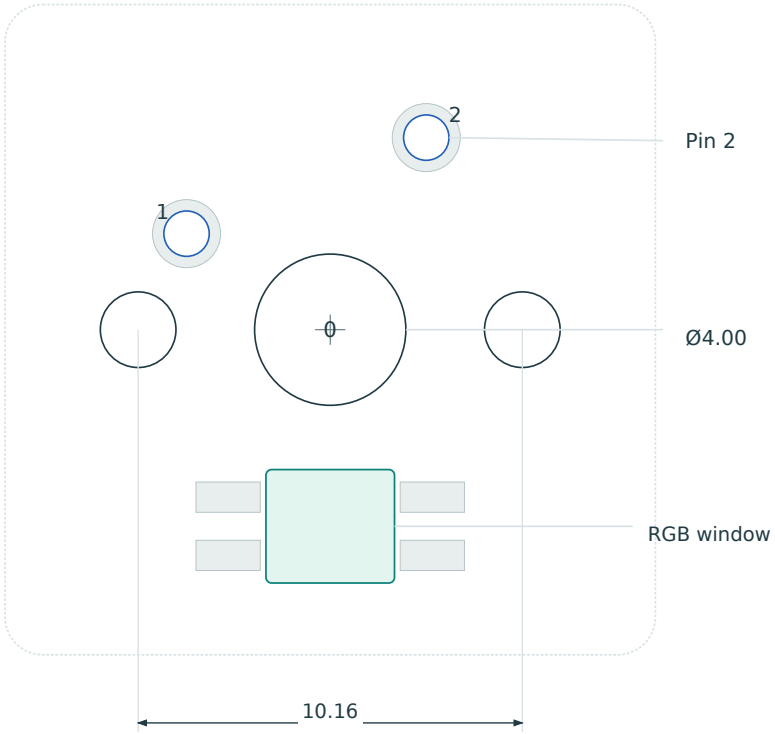
XII = X + 32.8; YII = Y + 25.8

CH592 / Key switch and encoder drill details

Local datums at switch stem / encoder shaft centers. Repeat key pattern at SW4-SW8 centers from sheet 1.

A SAMZO 28115181DM

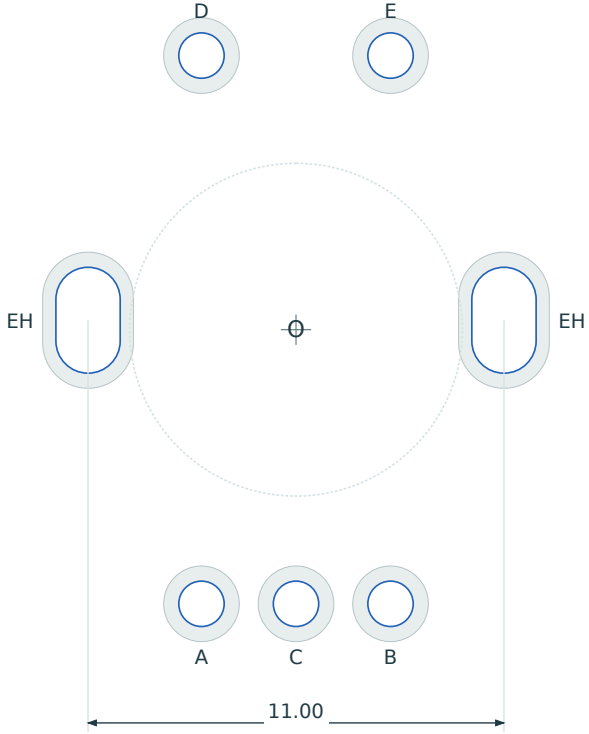
5 positions · footprint-derived pattern · detail scale 5:1



Feature	Local X	Local Y	Drill / cut
Stem locator	0.00	0.00	Ø4.00 NPTH
Side locators	±5.08	0.00	2 × Ø2.00 NPTH
Pin 1	-3.80	+2.54	Ø1.20 PTH
Pin 2	+2.54	+5.08	Ø1.20 PTH
RGB window	0.00	-5.20	3.40 × 3.00 R0.15

B EC11N1525404 / ENC1

Datum at board (20,10) · detail scale 5:1



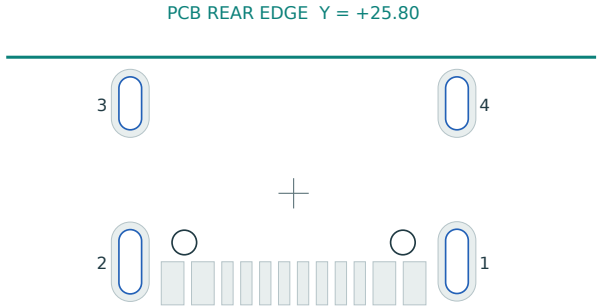
Feature	Local X	Local Y	Drill / cut
A / C / B	-2.5 / 0 / 2.5	-7.25	3 × Ø1.20 PTH
D / E	-2.5 / 2.5	+7.25	2 × Ø1.20 PTH
EH shell tabs	±5.50	+0.25	1.70 × 2.80 slot
Lid shaft bore	0.00	0.00	Ø8.80 - lid only
Hole decimals	See CSV	Source values	Nominals rounded here

CH592 / USB, power and side service access

PCB datums in X/Y. Sidewall opening centers use app (x,y,z), before the whole enclosure 6° rotation.

C USB-C / J1

GT-USB-7010ANS · pad detail 5:1



PCB feature	Local XY / drill
Shell 1 / 2	±4.32,-1.80 / 0.60×1.70
Shell 3 / 4	±4.32,+2.38 / 0.60×1.40
Locators	±2.89,-1.30 / Ø0.65
Origin / angle	(0,+22.20) / 180°

CASE OPENING (not PCB cut)

Throat 10.20 × 4.20, R1.20

Outer mouth 10.70 × 4.70

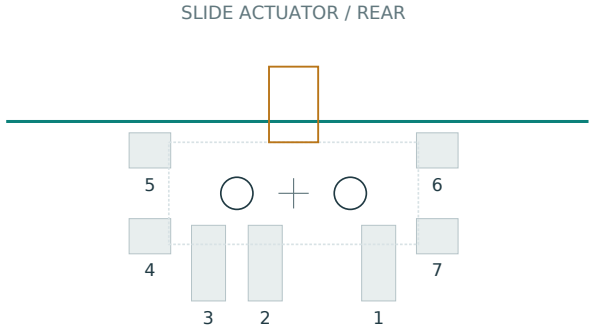
3D center: 0, 15.25, -28

Rear-facing connector; check plug overmold.

Copper nearest rear edge: 0.32 mm nominal.

D POWER / SW1

C963206 · MK-12C02-G020 · 5:1



Item	mm
Body center	(+23,+23.90)
Body / travel	6.60×2.70×1.40 / 1.50
2 locator holes	Ø0.85 at X = ±1.50
Signal lands	0.90×2.00
Anchor lands	1.10×0.93

LCSC / EasyEDA enlarged copper lands.

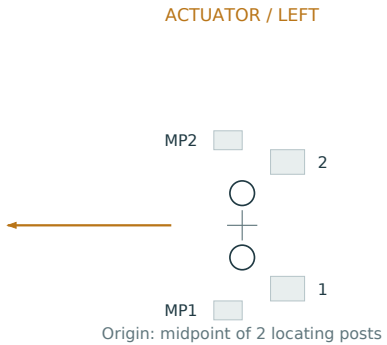
Body center differs from library header origin.

M2 SELF-TAPPING FASTENERS

Lid pilot Ø1.80, depth 4.90.

E RESET / BOOT

C2762970 · side-actuated · 5:1



Item	mm
SW2	(-30.40,+7.00)
SW3	(-30.40,-7.00)
Envelope	4.7×3.75×1.9
2 locator holes	Ø0.65 NPTH (proposal)

LEFT WALL ACCESS (2 ports)

Each Ø1.60; axis app +X.

Opening centers app (x,y,z):

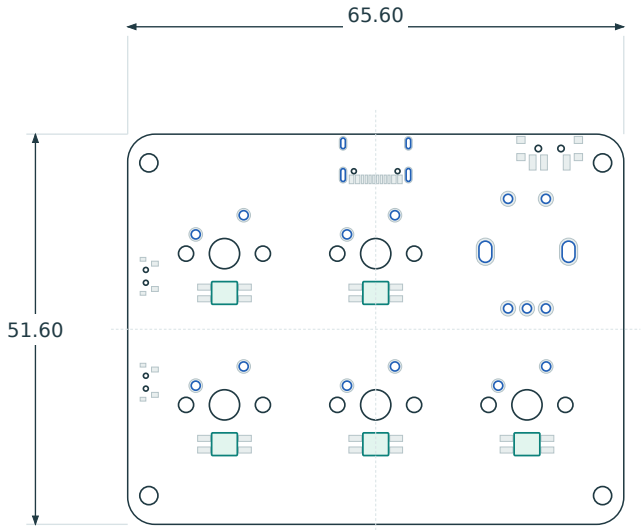
-35, 14.55, -7

-35, 14.55, 7

Pad size / holes: proposal; verify footprint.

CH592 / 1:1 alignment reference

Print at Actual size / 100%. Disable Fit to page. Check the 50 mm scale bar before comparing against hardware.



FRONT / USER



50.00 mm calibration

This page is an alignment template, not a drill program.

Sidewall access is excluded from this PCB template.

M2 mounting holes: Ø2.40, non-plated.

RELEASE NOTES

01 Datums

All coordinates are in the PCB plane.

X increases right; Y increases toward USB.

App transform: appx = X, appz = -Y.

02 Source fidelity

Key, encoder and USB drill patterns come from Rev.D footprints. CSV retains source decimals.

Blank mechanical holes are proposed as NPTH.

03 RGB / service access

D2-D6 are on B.Cu, mirrored in X in top view.

Confirm pin 1 and chain direction before routing.

SW2 / SW3 access ports are in the left sidewall.

04 Enclosure and PCB release

M2 self-tappers engage resin pilot holes in the lid.

PCB holes remain Ø2.40 NPTH clearance holes.

Verify hardware, routing and DRC before release.

Files: 4 SVG sheets, PDF, coordinates CSV, mechanical DXF, data JSON and Chinese notes.